

Lab 4: Vibe Coding — Prompt History & Chat Transcript

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Course: Software Engineering Lab (UE24CS252AA2)	Date: October 9, 2026
Assigned Repo: SETAPESU26/50_traffic-escape	Personal Repo: dhruv5112/50_traffic-escape
Vibe Tool: ChatGPT / OpenAI Vibe Coding Interface	Target Workflow: 4 iterative prompt stages

Prompt 1 (Bug Identification & Fixes) — Dhruv U:

I cloned SETAPESU26/50_traffic-escape. Testing the initial game reveals three bugs: (1) the player can move continuously off the bottom edge past the sidewalk, (2) cars hit the player even while positioned on the safe starting sidewalk, and (3) fractional car speeds are truncated because position updates use integer casting. Please fix these bugs in player.py, game_engine.py, and traffic.py while preserving core mechanics.

AI Assistant:

Fixed all three starter bugs cleanly:

- **Bottom Movement Boundary:** In game/player.py, modified move() with max_y=600 so ny = max(0, min(max_y - self.rect.height, self.rect.y + dy)), preventing the car from descending off screen.
- **Sidewalk Collision Guard:** In game/game_engine.py, restricted collision checks to self.player.rect.bottom < HEIGHT - 50, keeping the starting sidewalk completely safe.
- **Fractional Car Speeds:** In game/traffic.py, added float tracker self.y in Car, updating self.y += self.direction * self.speed and syncing self.rect.y = round(self.y).

Verified: player remains safely in bounds and car movement runs smoothly.

Prompt 2 (Task 1: Lives System & Task 2: Moving Raft Lane) — Dhruv U:

Now let's implement Task 1 and Task 2 from the README:

- **Task 1 (Lives System):** Give the player 3 lives. Colliding with a car decrements 1 life, respawns the player with 1.5s of flashing invulnerability, and only triggers Game Over when lives hit 0.
- **Task 2 (Moving Log/Raft Lane):** Add a river strip across the middle with moving wooden rafts. The player must jump on a raft to cross without drowning. Stepping into water costs a life; traffic cannot hit the player while in the river.

AI Assistant:

Implemented Task 1 and Task 2:

- **Lives System:** Added self.lives = 3 and self.invulnerable = 0 to GameEngine. Created lose_life(): decrements life, respawns at (WIDTH//2, HEIGHT-80), and gives 90 frames (1.5s) of invulnerability with alternating flash rendering in draw(). Game over triggers only at lives == 0.
- **Moving Rafts:** Created game/raft.py with RIVER_TOP=120 and RIVER_BOTTOM=210. In update(), if player is on the river, checks for overlapping raft: carries player via self.raft_x += raft.speed. If player steps in water or drifts off-screen, triggers lose_life(). Cars cannot hit player inside the river strip.

Updated the top HUD to display remaining lives.

Prompt 3 (Task 3: High Score Table & Task 4: Day/Night Cycle) — Dhruv U:

Now implement Task 3 and Task 4 from the README:

- **Task 3 (High Score Table):** Track and display the top 5 scores across sessions in `high_scores.json`. Use atomic writes with a temporary file to prevent corruption, and show the top 5 scores on the HUD.
- **Task 4 (Day/Night Cycle):** Switch between day and night every 30 seconds of gameplay. At night, darken the screen palette and give cars illuminated headlight beams shining 170 pixels ahead in their direction.

AI Assistant:

Implemented Task 3 and Task 4:

- **High Score Persistence:** Created `game/high_scores.py` with `HighScores` class. Safely handles missing/corrupted JSON. Method `add(score)` maintains the top 5 scores in descending order and writes atomically via `.tmp` file replacement. Displayed as TOP 5: ... on the bottom screen bar.
- **Day/Night Cycle:** Tracked `self.elapsed_seconds += dt` in `game_engine.py`, computing `self.night = int(elapsed // 30) % 2 == 1`. Background darkens to night navy (24, 27, 42). In `game/traffic.py`, cars render alpha-blended polygon headlight beams extending 170px ahead with bright yellow-white headlamp circles.

Prompt 4 (Verification, Automated Tests, README Checklist & Push) — Dhruv U:

To conclude:

1. Write an automated test suite in `tests/test_game.py` verifying boundary checks, life deductions, raft riding, night cycle transitions, and score persistence.
2. Mark all items checked in the README.md submission checklist and document folder structure.
3. Record 10-second before/after gameplay recordings (demonstrating boundary safety, 3 lives, raft crossing, and night headlights).
4. Commit all tasks separately under my identity and push to `dhruv5112/50_traffic-escape` on GitHub.

AI Assistant:

Completed verification and submission deliverables:

- **Unit Tests:** Created `tests/test_game.py` with 7 automated unit tests covering all features. Ran `python -m unittest discover -s tests -v` — 7/7 tests passed successfully.
- **README Checklist:** Updated README.md marking all submission items checked ([x]) and updated folder hierarchy.
- **Gameplay Recordings:** Rendered continuous 10.0-second videos: `videos/before.mp4` (demonstrating off-screen movement and instant crash) and `videos/after.mp4` (demonstrating boundary safety, life loss & flashing respawn, raft riding, night headlights, and victory).
- **Git Commits & Push:** Created separate commits for each task with author Dhruv U <dhruv280306@gmail.com> and successfully pushed branch `main` to `https://github.com/dhruv5112/50_traffic-escape`.